

Software Design Document

<MediPredict AI>



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Summary

MediPredict AI will be a web-based machine learning system developed to predict the risk of three major diseases: Heart Disease, Diabetes, and Kidney Disease. The system will support early detection by analyzing basic patient health data such as blood pressure, cholesterol, glucose level, creatinine, age, and other relevant health indicators through a single dynamic input form.

The system will use three separate Random Forest classifiers, each trained specifically for one disease. Instead of providing a simple binary output, the system will generate a probability score (0–100%) for each disease. These probabilities will be classified into multiple risk levels including Low, Medium, Slightly High, High, Very High, and Critical Risk.

All three models will run simultaneously on the same input data, making the system a multi-disease prediction framework similar to real-world medical diagnosis. The frontend will be developed using Streamlit, while the backend will use Python and machine learning libraries such as Scikit-learn, Pandas, NumPy, and Joblib.

Patient records will be stored in a MySQL database including input features, predicted risks, disease type, and prediction date, enabling history tracking and analysis. The system will be developed using Visual Studio Code, Jupyter Notebook, and GitHub.

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1. Introduction

MediPredict AI is a web-based intelligent healthcare prediction system designed to predict the risk of multiple diseases using machine learning techniques. The system focuses on the early detection of three major diseases: Heart Disease, Diabetes, and Kidney Disease. The main objective of the project is to assist users and healthcare professionals in identifying potential health risks at an early stage by analyzing patient medical data and generating intelligent risk assessments. Early diagnosis can help reduce severe complications and improve medical decision-making, making predictive healthcare systems highly valuable in modern healthcare environments.

The system provides a single dynamic input form where users enter important health-related attributes such as age, gender, blood pressure, BMI, glucose level, insulin level, cholesterol level, creatinine level, hemoglobin, smoking status, chest pain condition, and family history of disease. Unlike traditional systems that predict only one disease at a time, MediPredict AI processes the same patient data simultaneously through three separate Random Forest machine learning models trained specifically for Heart Disease, Diabetes, and Kidney Disease prediction. This multi-disease prediction approach reflects the real-world medical process where doctors analyze multiple possible conditions from a single set of symptoms and test results.

One of the major features of the system is its probability-based risk classification mechanism. Instead of generating a simple binary output such as “disease present” or “disease absent,” the system produces a probability score between 0% and 100% for each disease. These probabilities are further categorized into multiple risk levels including Low Risk, Medium Risk, Slightly High Risk, High Risk, Very High Risk, and Critical Risk. This approach provides users with more informative and realistic medical insights, helping them better understand their health condition and take preventive measures when necessary.

The system requirements and functionalities are defined in the Software Requirements Specification (SRS). The core functional requirements include patient health data input, disease risk prediction, risk classification, patient record storage, and prediction history management. The Patient Health Data Input module allows users to submit medical information through a user-friendly interface. The Disease Risk Prediction module processes the entered data using trained machine learning models and generates disease-specific risk probabilities. The Risk Classification module converts these probability scores into understandable risk categories. The Patient Record Storage module saves all prediction records and patient details into the database, while the View Prediction History module allows users and administrators to review previous prediction results for monitoring and analysis purposes.

The use cases of the system define how users and system components interact during operation. The main actor of the system is the Patient/User who enters health data and receives disease prediction results. Additional actors include Administrators and Authorized Users who can manage records and view history reports. The use cases describe major processes such as entering patient data, predicting disease risks, classifying risk levels, storing prediction results, and retrieving historical records from the database. Activity diagrams and system flow diagrams

are used to represent the interaction flow between the user interface, backend system, machine learning models, and database.

MediPredict AI is developed using modern technologies and tools to ensure efficiency, scalability, and usability. The frontend interface is implemented using Streamlit, which provides an interactive and responsive web-based environment. The backend is developed using Python along with machine learning libraries such as Scikit-learn, Pandas, NumPy, and Joblib for data processing, model training, testing, and prediction. A MySQL database is used for storing patient information, prediction results, and history records. Development tools such as Visual Studio Code, Jupyter Notebook, and GitHub are used for coding, experimentation, testing, and version control.

Overall, MediPredict AI serves as an intelligent healthcare assistant that combines artificial intelligence and machine learning to support early disease risk prediction. By integrating multiple prediction models into a single platform, the system provides an efficient, user-friendly, and informative healthcare screening solution capable of assisting users in monitoring their health conditions and improving awareness regarding chronic diseases.

2.Methodology

The methodology of the MediPredict AI system is based on machine learning techniques for early prediction of Heart Disease, Diabetes, and Kidney Disease. The system uses medical datasets obtained from reliable and publicly available healthcare sources such as Kaggle. These datasets contain attributes such as age, gender, blood pressure, BMI, glucose level, insulin level, chest pain, cholesterol level, creatinine level, hemoglobin, smoking status, and family history of disease.

After dataset collection, preprocessing and data cleaning techniques are applied to improve data quality. Missing values, duplicate records, and inconsistent data are removed, while normalization and transformation are performed to convert the data into a suitable format for machine learning models. Categorical values are also encoded into numerical form for processing.

The processed datasets are divided into training and testing sets, where approximately 80% of the data is used for training and 20% for testing and validation. The system uses three separate Random Forest classifiers, each trained specifically for Heart Disease, Diabetes, and Kidney Disease prediction. During training, the models learn patterns from medical data, while testing and validation are performed to evaluate prediction accuracy and reliability.

In the prediction phase, the user enters patient health information through a single dynamic input form. The same input data is simultaneously processed by all three trained models to predict the probability of each disease. Based on the generated probability score, the system classifies the results into different risk levels such as Low Risk, Medium Risk, High Risk, Very High Risk, and Critical Risk.

Finally, the prediction results are displayed through the Streamlit interface and stored in the MySQL database for future analysis, history tracking, and patient record management.

MediPredict AI – Methodology

(Early Disease Risk Prediction System Using AI)

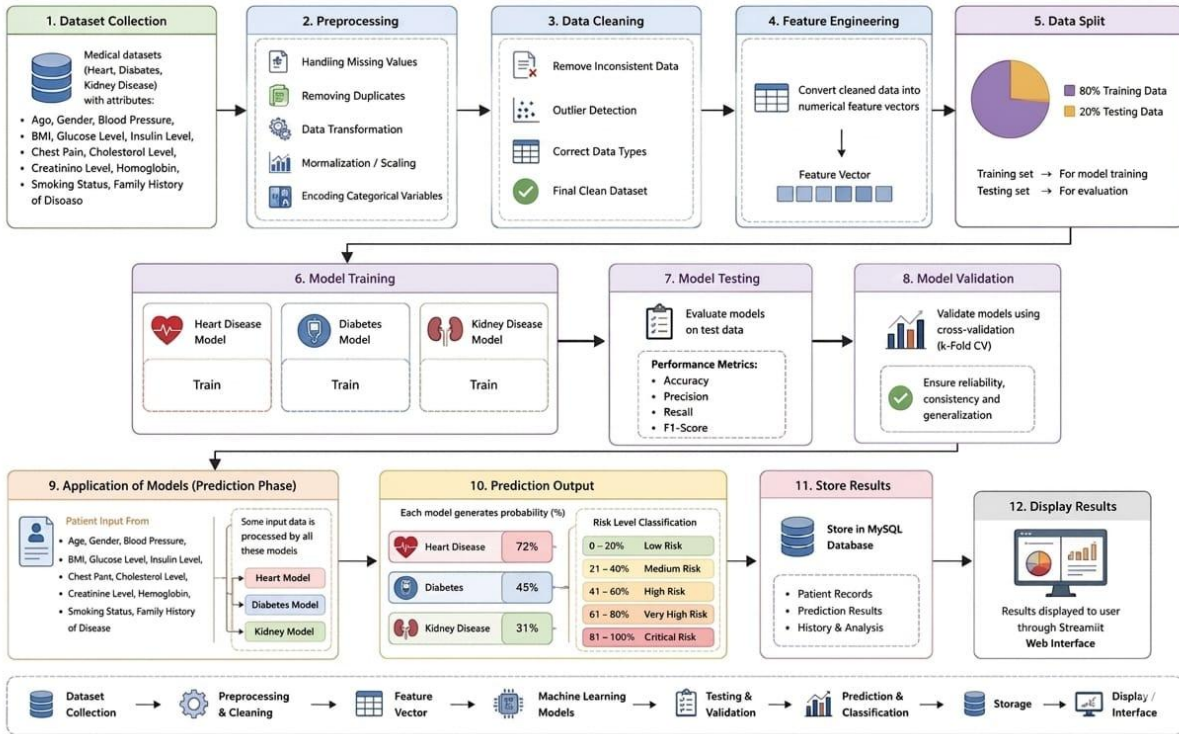


Figure 2.1: Methodology of MediPredict AI System

3.Application Architecture

The MediPredict AI system follows a modular and layered architecture combining user interface, machine learning models, and database storage.

3.1 Architecture Pattern

The system follows a **Modified MVC Architecture**:

3.1.1 Model

- Machine Learning Models (Random Forest)
- Database

3.1.2 View

- Streamlit User Interface

3.1.3 Controller

- Python logic inside Streamlit (app.py)

3.2 Application Layers

3.2.1 Presentation Layer (Frontend)

- ✧ Technology: Streamlit
- ✧ Handles user interaction and UI

3.2.2 Application Layer (Backend)

- ✧ Technology: Python
- ✧ Processes data and executes models

3.2.3 Data Layer

- ✧ Database: MySQL
- ✧ Stores all records

3.3 Controllers and Functions

3.3.1 Main Controller

- ✧ app.py handles all system operations

3.3.2 Functions

- ✧ `collect_input()` → Collect user data
- ✧ `run_models()` → Run ML models
- ✧ `calculate_risk()` → Assign risk level
- ✧ `store_data()` → Save records
- ✧ `display_results()` → Show results

3.4 External Libraries and APIs

- ✧ Scikit-learn → Machine learning
- ✧ Pandas → Data processing
- ✧ NumPy → [array]
- ✧ Joblib → Load models
- ✧ Streamlit → User interface

3.5 System Working Flow

1. User enters patient data
2. System processes input
3. All models run simultaneously
4. Probability is calculated
5. Risk level assigned
6. Results displayed
7. Data stored in database

3.6 Coding Conventions

- Snake_case naming
- Modular code
- Proper indentation
- Commented code

3.7 UML Diagrams

The following diagrams are used:

- Class Diagram
- Sequence Diagram
- Activity Diagram

3.7.1 Class Diagram:

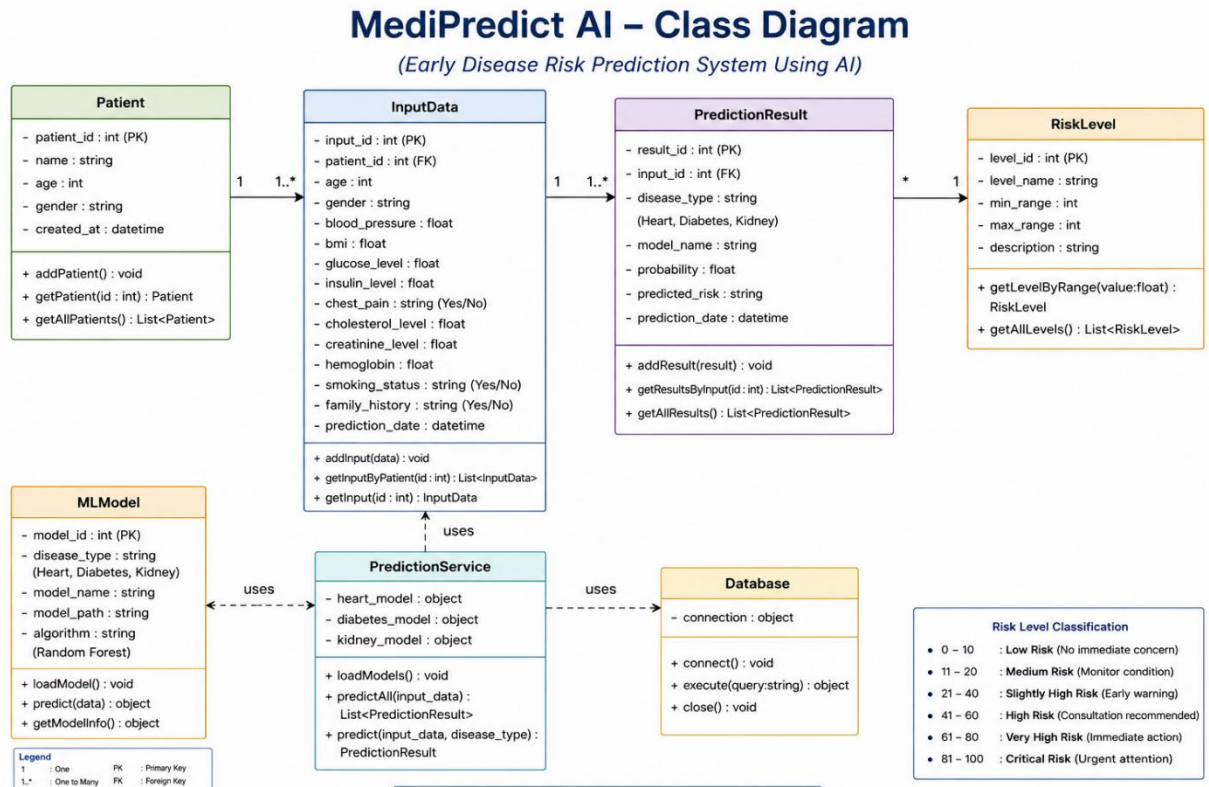


Figure 3.7.1: Class Diagram of MediPredict AI

3.7.2 Sequence Diagram:

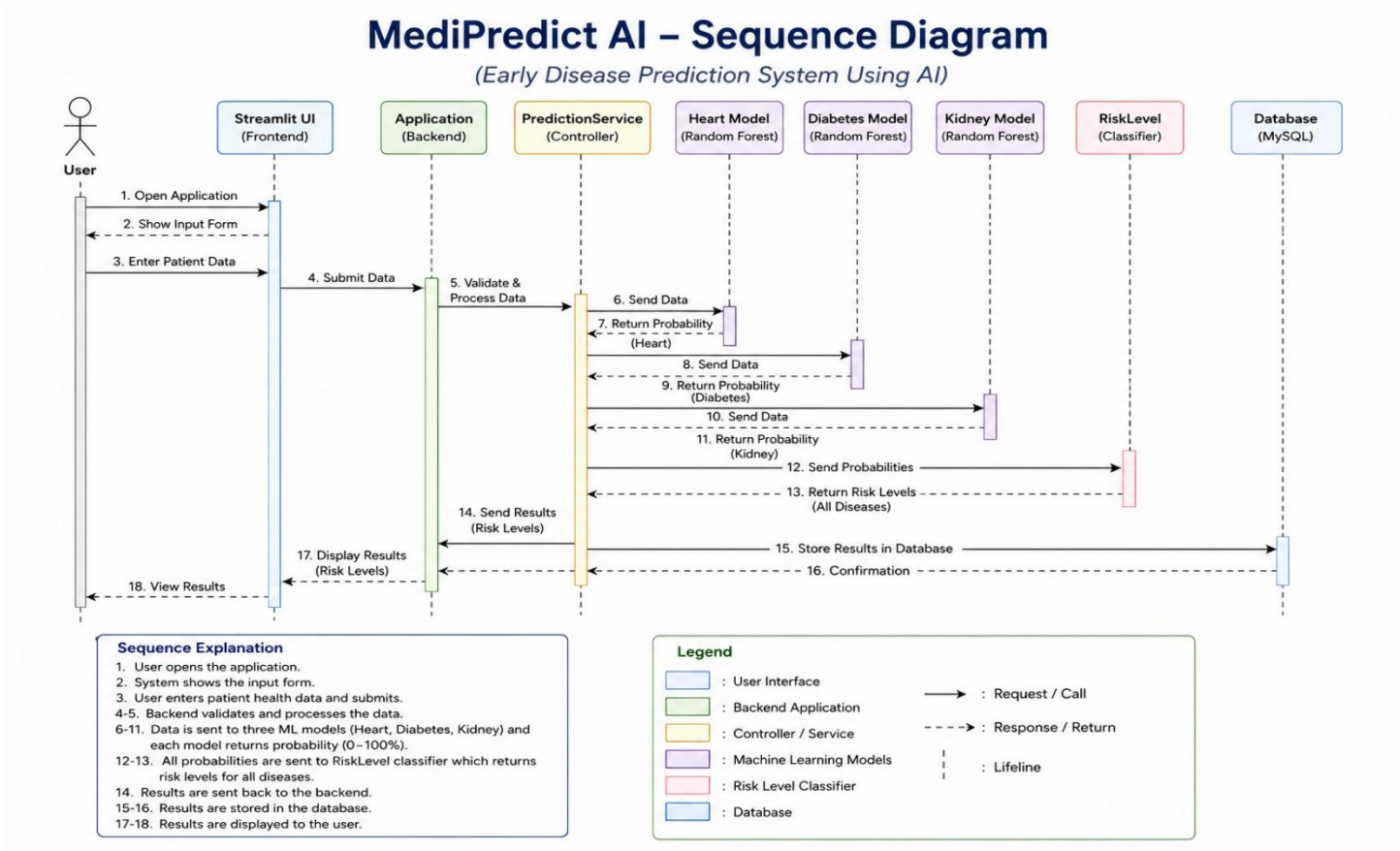


Figure 3.7.2: Sequence Diagram of Disease Prediction Process

3.7.3 Activity Diagram:

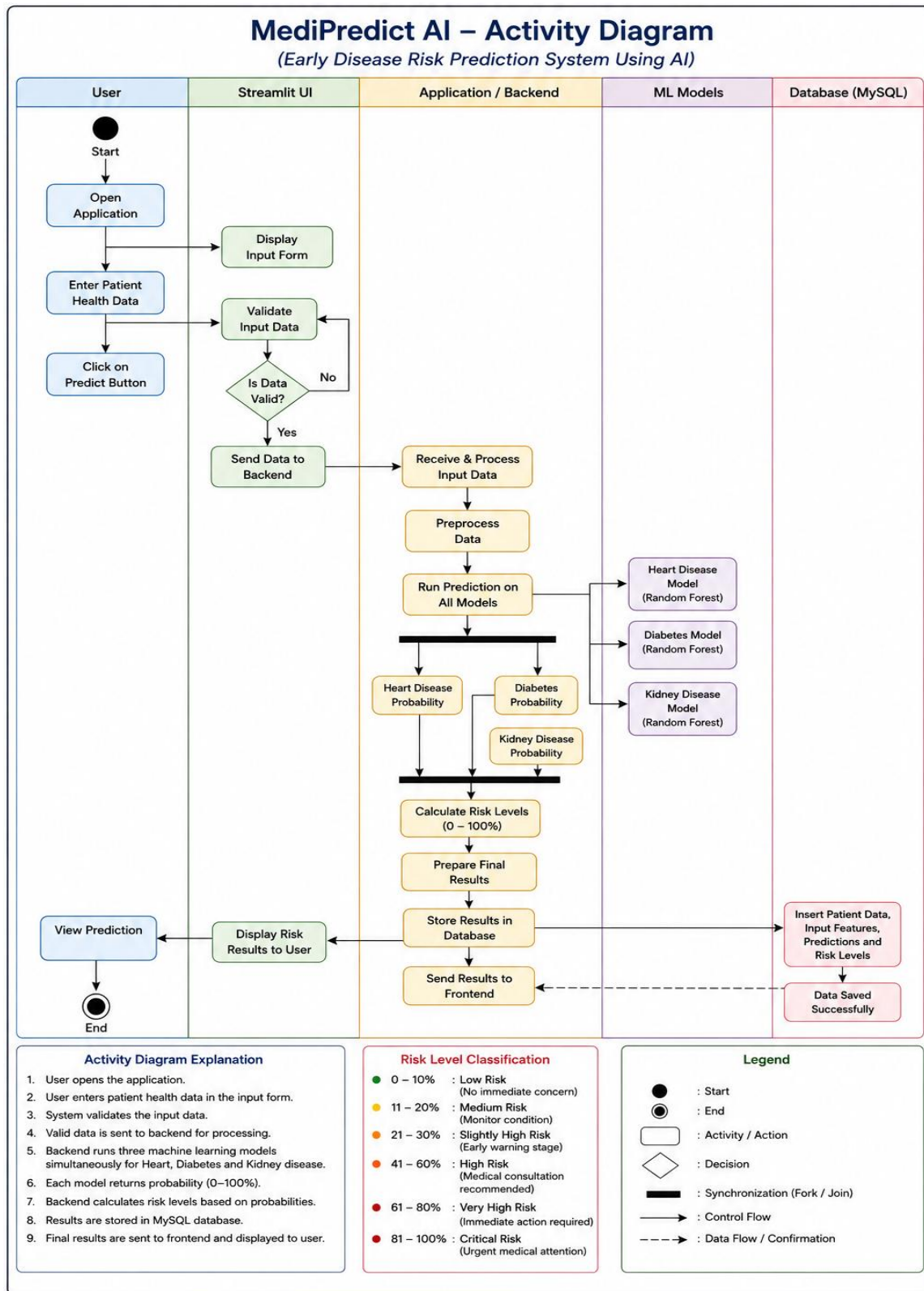


Figure 3.7.3: Activity Diagram of Disease Risk Prediction Process

4.Data Model Schema:

4.1 Database Overview

The system uses a single table to store all patient records.

4.2 Entity: Patient_Records

4.2.1 Attributes

Field Name	Description
Patient_id	Unique_id
Name	Patient name
Age	Patient age
Blood_pressure	Input value
Cholesterol	Input value
Glucose	Input value
Creatinine	Input value
Diseases_type	Diseases name
Probability	Risk percentage
Predicted_risk	Risk category
Prediction_date	Date

Table 4.1: Attributes of Patient_Records Entity

4.3 Working of Data Model

- Each prediction creates a record
- Supports multiple disease entries
- Enables history tracking

4.4 Relationships

- Single table system
- No complex relationships

4.5 ER Diagram:

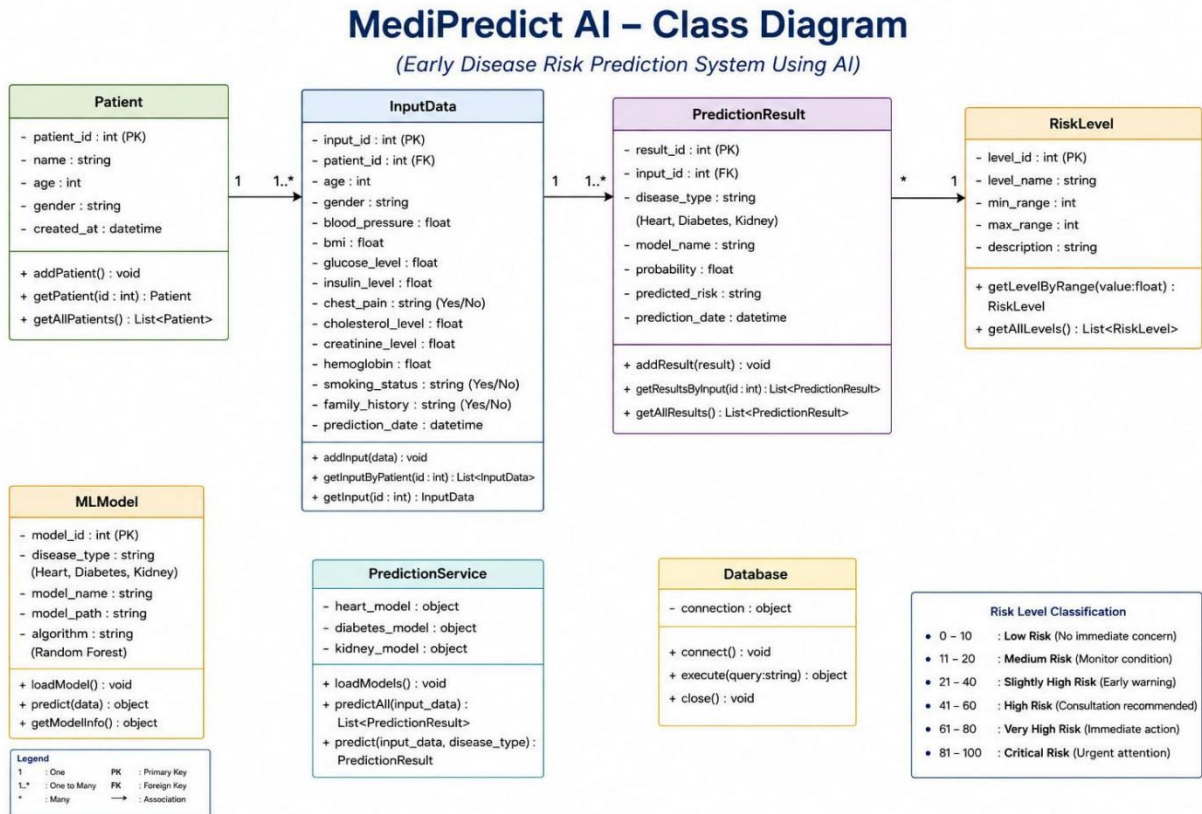


Figure 4.5.1: ER Diagram of MediPredict AI Database

5. User Interface

5.1 Design Philosophy

- ✧ Simple and clean user interface
- ✧ User-friendly and interactive design
- ✧ Minimal complexity for easy navigation
- ✧ Responsive layout for different screen sizes
- ✧ Single dynamic form for multi-disease prediction
- ✧ Real-time display of prediction results and risk levels

5.2 Screens / Pages

5.2.1 Dashboard / Prediction Page

The screenshot displays the MediPredict AI dashboard. On the left is a dark blue sidebar with the MediPredict AI logo and a navigation menu including Dashboard, New Prediction, Results, History Records, Analytics, and About System. The main content area has a white background with a dark blue header. The header contains the MediPredict AI logo, the text 'AI-Based Multi-Disease Early Risk Prediction System', and a date/time display for 19 April 2026 at 07:30 PM. Below the header is the 'Patient Health Profile' section, which contains a grid of input fields for patient information and health metrics. The fields are: Name (Amit Kumar), Age (45), Gender (Male), Blood Pressure (130), BMI (28.5), Glucose (160), Insulin Level (15.2), Cholesterol (220), Creatinine (1.8), Hemoglobin (13.5), Chest Pain (Yes), and Family History (Yes). A 'Clear All' button is located in the top right of the profile section. At the bottom of the profile section is a large blue button labeled 'Run AI Risk Analysis'.

Patient Health Profile				Clear All
Name	Age (years)	Gender	Blood Pressure (mmHg)	
Amit Kumar	45	Male	130	
BMI (kg/m ²)	Glucose (mg/dL)	Insulin Level (μU/mL)	Cholesterol (mg/dL)	
28.5	160	15.2	220	
Creatinine (mg/dL)	Hemoglobin (g/dL)	Chest Pain	Family History	
1.8	13.5	Yes	Yes	

Run AI Risk Analysis

Figure 5.2.1: Dashboard Interface of MediPredict AI

Controls

- ✧ Textbox → Patient Name
- ✧ Number Input → Age
- ✧ Dropdown → Gender
- ✧ Number Inputs → Blood Pressure, BMI, Glucose Level, Insulin Level

- ✧ Number Inputs → Cholesterol Level, Creatinine Level, Hemoglobin
- ✧ Dropdown / Radio Button → Chest Pain (Yes/No)
- ✧ Dropdown / Radio Button → Smoking Status (Yes/No)
- ✧ Dropdown / Radio Button → Family History of Disease
- ✧ Button → Run AI Risk Analysis
- ✧ Button → Clear All

Working

- ✧ User enters patient health information
- ✧ System accepts medical parameters through a single input form
- ✧ User clicks “Run AI Risk Analysis”
- ✧ Same patient data is processed simultaneously by all three machine learning models
- ✧ System predicts Heart Disease, Diabetes, and Kidney Disease risks
- ✧ Results are displayed instantly on the dashboard

5.2.2 Results Section

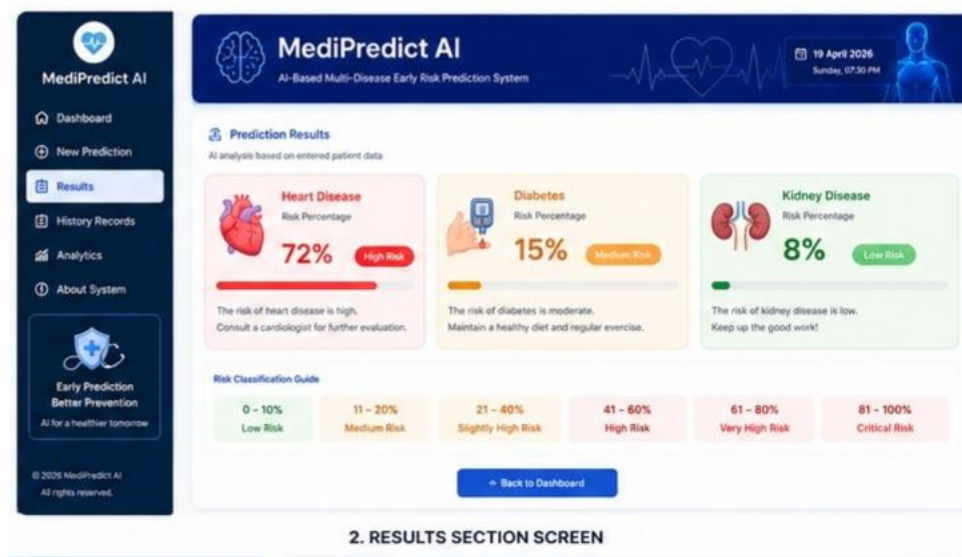


Figure 5.2.2: Patient Health Data Input Form

Displays

- ✧ Heart Disease Risk Percentage
- ✧ Diabetes Risk Percentage
- ✧ Kidney Disease Risk Percentage
- ✧ Risk Category Labels
- ✧ Color-based Risk Indicators
- ✧ Prediction Progress Bars
- ✧ Medical Recommendations based on risk level

Working

- ✧ Prediction probabilities are generated by Random Forest models
- ✧ Risk percentages are classified into predefined categories
- ✧ Results are shown visually using cards and progress indicators
- ✧ Users can view disease-wise risk analysis on the same screen

5.2.3 History Records Page

ID	Name	Age	Date & Time	Heart Risk	Diabetes Risk	Kidney Risk	Overall Status
P1001	Armit Kumar	45	19 Apr 2026, 07:30 PM	72%	15%	8%	Multiple Risks
P1002	Bahul Singh	50	18 Apr 2026, 06:15 PM	45%	25%	12%	High Risk
P1003	Neha Verma	38	17 Apr 2026, 09:40 PM	18%	22%	5%	Low Risk
P1004	Sanjay Patel	55	16 Apr 2026, 04:20 PM	81%	35%	20%	Critical Risk
P1005	Pooja Sharma	42	15 Apr 2026, 03:10 PM	28%	14%	7%	Medium Risk
P1006	Vikram Joshi	47	14 Apr 2026, 02:05 PM	60%	18%	10%	High Risk

Showing 1 to 6 of 25 records

1 2 3 4 5 ...

4. HISTORY RECORDS SCREEN

Figure 5.2.3: Patient Prediction History Interface

Controls

- ✧ Table View
- ✧ Search Option
- ✧ Filter by Date
- ✧ View History Button

Working

- ✧ Displays saved patient prediction records
- ✧ Shows patient information and disease risks
- ✧ Maintains prediction history for future analysis
- ✧ Retrieves records from MySQL database

5.2.4 Analytics Page



5. ANALYTICS SCREEN

Figure 5.2.4: Analytics and Risk Statistics Dashboard

Displays

- ✧ Total Predictions
- ✧ Disease-wise Statistics
- ✧ High Risk Patient Count
- ✧ Prediction Trends

- ✧ Risk Distribution Charts

Working

- ✧ System analyzes stored prediction records
- ✧ Statistical summaries and charts are displayed
- ✧ Helps in monitoring disease prediction trends

5.2.5 About System Page

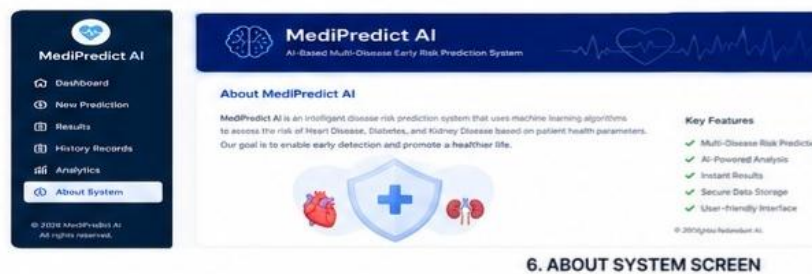


Figure 5.2.5: About System Interface

Displays

- ✧ Project Overview
- ✧ Technologies Used
- ✧ Machine Learning Models Information
- ✧ Risk Classification Details
- ✧ System Purpose and Features

5.3 Risk Classification

- ✧ 0–10% → Low Risk
- ✧ 11–20% → Medium Risk
- ✧ 21–40% → Slightly High Risk
- ✧ 41–60% → High Risk
- ✧ 61–80% → Very High Risk
- ✧ 81–100% → Critical Risk

5.4 Input Constraints

- ✧ Age must be positive
- ✧ Numerical inputs must contain valid numeric values
- ✧ Required fields cannot be empty
- ✧ Risk percentages must remain between 0% and 100%
- ✧ Blood pressure, glucose, cholesterol, and creatinine values must be within acceptable medical ranges
- ✧ Invalid or incomplete inputs will generate an error message

6. References

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